

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Expt 6</b> | <b>Finite Word length Effects</b>                                                                                 |
| <b>Date:</b>  | <b>6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB</b> |

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Signal Parameters
```

```
fs = 1000;      % Sampling frequency (Hz)
```

```
f = 50;         % Signal frequency (Hz)
```

```
t = 0:1/fs:0.1; % Time vector
```

```
x = sin(2*pi*f*t); % Original sine wave
```

```
%% Quantization Levels
```

```
L = [8 16 32];
```

```
colors = {'r','g','m'};
```

```
mse = zeros(1,3);
```

```
xq = cell(1,3);
```

```
e = cell(1,3);
```

```
%% Quantization
```

```
figure;
```

```
% Original Signal
```

```
subplot(4,1,1);
```

```
plot(t,x,'LineWidth',1.5);
```

```

title('Original Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

for i = 1:length(L)

    % Uniform Quantization
    xq{i} = round((x+1)*(L(i)-1)/2) * (2/(L(i)-1)) - 1;

    % Quantization Error
    e{i} = x - xq{i};

    % Mean Square Error
    mse(i) = mean(e{i}.^2);

    % Plot Quantized Signal
    subplot(4,1,i+1);
    stairs(t,xq{i},colors{i},'LineWidth',1.2);
    title(['Quantized Signal ( ' num2str(L(i)) ' Levels)']);
    xlabel('Time (s)');
    ylabel('Amplitude');
    grid on;
end

%% Display MSE
fprintf('MSE for 8 Levels = %f\n', mse(1));
fprintf('MSE for 16 Levels = %f\n', mse(2));

```

```
fprintf('MSE for 32 Levels = %f\n', mse(3));
```

```
%% Error Plots
```

```
figure;
```

```
for i = 1:3
```

```
    subplot(3,1,i);
```

```
    plot(t,e{i},'LineWidth',1.2);
```

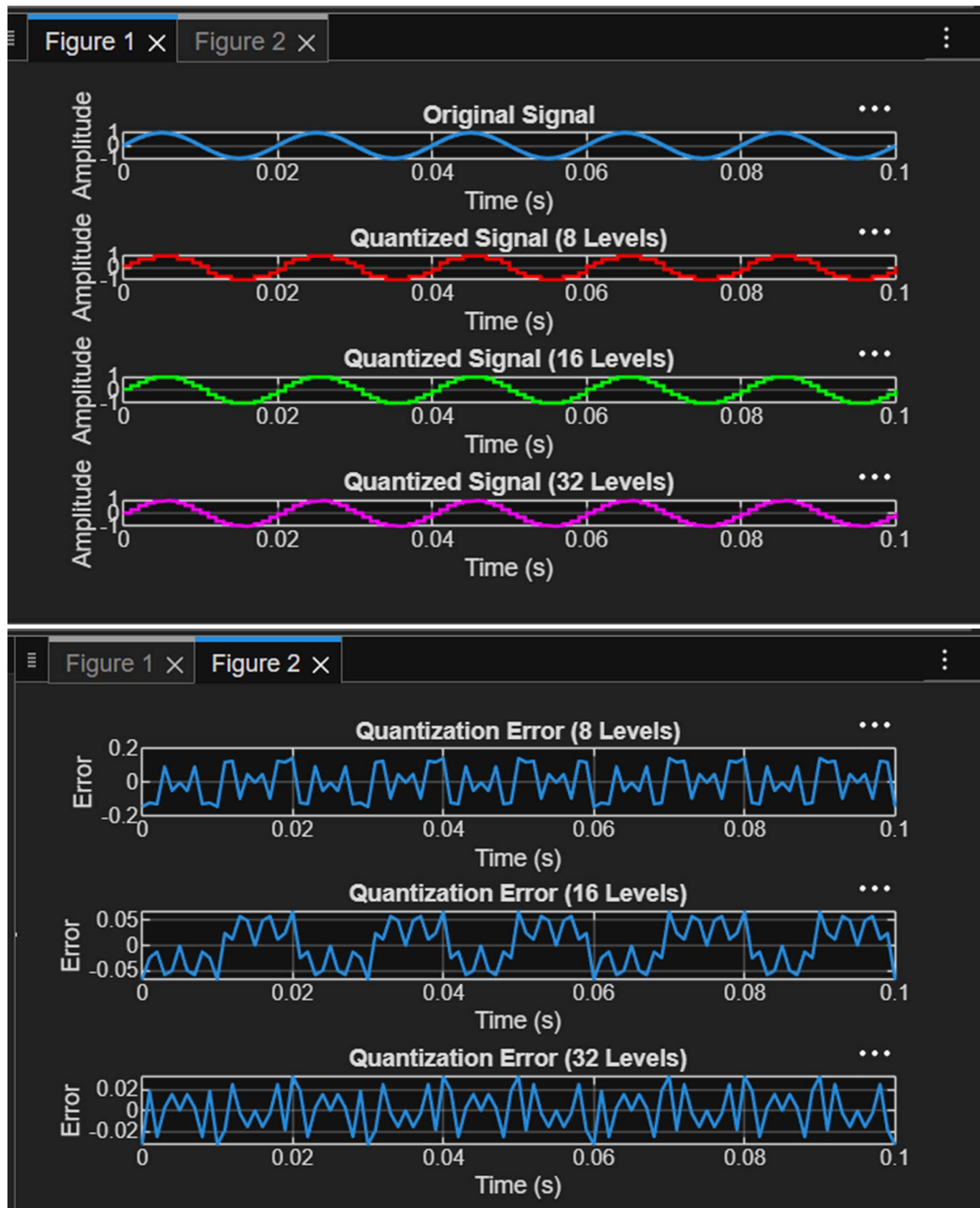
```
    title(['Quantization Error ( ' num2str(L(i)) ' Levels)']);
```

```
    xlabel('Time (s)');
```

```
    ylabel('Error');
```

```
    grid on;
```

```
end
```



Result: 1. A discrete-time sinusoidal signal was successfully generated in MATLAB.

2. Uniform quantization was performed for different quantization levels, and the corresponding quantized signals were obtained.

3. Comparison of the original and quantized signals showed that the quantized signal approximates the original signal more closely as the number of quantization levels increases.

4. The quantization error was analyzed and found to decrease with increasing quantization levels, resulting in improved signal quality and reduced quantization noise.